

ELASTIC DIFFUSIVE COSMOLOGY

BOOK II

Emergent Gravity from Plenum Dynamics

Mathematical Derivations with Full Rigor
Every Step Documented — Every Dimension Checked

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Physical Insights and Theory Development

with Claude (Anthropic)

Mathematical Verification and Documentation

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“Gravitacija nije sila — to je tok.”

“Gravity is not a force — it is a flow.”

“Bez grešaka i pretpostavki.”

“Without errors and assumptions.”

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Preface

This book presents the mathematical foundations of Elastic Diffusive Cosmology (EDC), focusing on the emergent nature of gravity from the dynamics of the Plenum — the 5-dimensional energetic fluid that fills the Bulk in which our 3-dimensional membrane universe is embedded.

Purpose of This Book

The primary purpose of Book II is to provide **complete mathematical rigor** for the gravitational sector of EDC. Every derivation is presented step-by-step, with:

- Explicit statement of assumptions
- Numbered equations for easy reference
- Dimensional checks at every critical step
- Clear epistemic classification (Derived, Identified, or Proposed)

Epistemic Classification System

Throughout this book, we use a rigorous classification system:

Code	Meaning	Confidence
[D]	Mathematically derived from explicit assumptions	High
[I]	Numerical match, not derived from first principles	Medium
[P]	Hypothesis requiring further investigation	Low
[BL]	Experimental value (CODATA)	High
[Cal]	Calibrated to match observations	—

This system ensures intellectual honesty and helps reviewers immediately identify the epistemological status of each claim.

Notation Convention

Critical Notation Warning

This book uses two different symbols for quantities with dimensions of mass:

- M — Gravitational source mass (e.g., in $v(r) = \sqrt{2GM/r}$)
- $\mathcal{M}_v$ — Equivalent mass of vortex elastic energy: $\mathcal{M}_v = \sigma r_e^2 / c^2$

Never confuse M and $\mathcal{M}_v$! They have the same dimensions but fundamentally different physical meanings.

Acknowledgments

This work represents a collaboration between human physical insight and AI mathematical verification. The physical ideas, interpretations, and theoretical framework are entirely human contributions. The AI assistant provided systematic verification, dimensional analysis, and documentation support.

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Chapter 1

Foundations and Notation

1.1 The EDC Framework

Elastic Diffusive Cosmology proposes that our observable universe is a 3-dimensional membrane (brane) embedded in a 5-dimensional space called the Bulk. The Bulk is filled with an energetic fluid called the **Plenum**.

Postulate 1.1 (Membrane Universe). The observable universe is a 3-dimensional hypersurface (membrane) embedded in a 5-dimensional Bulk space.

Postulate 1.2 (Plenum). The Bulk is filled with an energetic fluid called the Plenum, characterized by:

- Pressure p and density ρ related by a stiff equation of state: $p = \rho c^2$
- Effectively incompressible flow: $\nabla \cdot \vec{v} = 0$
- Very low viscosity in the relevant regime

Postulate 1.3 (Particles as Vortices). Elementary particles are topological defects (vortices) on the membrane, creating localized regions where Plenum pressure is completely excluded.

1.2 Symbol Definitions

1.2.1 Fundamental Parameters

Symbol	Name	Value	Dimensions	Status
c	Speed of light	2.998×10^8 m/s	$[L T^{-1}]$	[BL]
σ	Membrane tension	1.41×10^{18} J/m ²	$[M T^{-2}]$	[Cal]
r_e	Classical electron radius	2.818×10^{-15} m	$[L]$	[BL]
R_ξ	Compact dimension	2.16×10^{-18} m	$[L]$	[Cal]
m_e	Electron mass	9.109×10^{-31} kg	$[M]$	[BL]
G	Newton's constant	6.674×10^{-11} m ³ /(kg·s ²)	$[L^3 M^{-1} T^{-2}]$	[BL]
$\hbar$	Reduced Planck constant	1.055×10^{-34} J·s	$[M L^2 T^{-1}]$	[BL]
α	Fine structure constant	1/137.036	$[1]$	[BL]

Table 1.1: Fundamental parameters used in EDC derivations.

1.2.2 Derived Quantities

Symbol	Name	Definition	Dimensions
$\mathcal{E}_v$	Vortex elastic energy	σr_e^2	$[ML^2T^{-2}] = \text{J}$
$\mathcal{M}_v$	Vortex equivalent mass	$\mathcal{E}_v/c^2 = \sigma r_e^2/c^2$	$[M]$
r_{core}	Core radius	GM/c^2	$[L]$
r_s	Schwarzschild radius	$2GM/c^2$	$[L]$
ν_{bulk}	Kinematic viscosity	η/ρ	$[L^2T^{-1}]$

Table 1.2: Derived quantities and their definitions.

1.3 Dimensional Analysis Convention

Throughout this book, we verify every equation dimensionally using:

$$[L] = \text{length (meters)} \quad (1.1)$$

$$[M] = \text{mass (kilograms)} \quad (1.2)$$

$$[T] = \text{time (seconds)} \quad (1.3)$$

A dimensional check is denoted by $\checkmark$ when both sides match.

Chapter 2

Derivation of Plenum Flow Velocity

This chapter presents the complete derivation of the Plenum flow velocity profile around a gravitational source. This is the foundational result of EDC's gravitational sector.

2.1 Assumptions

Postulate 2.1 (Incompressibility). The Plenum is incompressible:

$$\nabla \cdot \vec{v} = 0 \quad (2.1)$$

Postulate 2.2 (Stationarity). The flow is stationary (time-independent):

$$\frac{\partial \vec{v}}{\partial t} = 0 \quad (2.2)$$

Postulate 2.3 (Inviscid Flow). The flow is inviscid in the relevant regime:

$$\eta \approx 0 \quad (2.3)$$

This assumption is justified in Chapter 4.

Postulate 2.4 (Pressure Exclusion). A vortex (particle) completely excludes Plenum pressure at its core:

$$p(r_{\text{core}}) = 0 \quad (2.4)$$

2.2 Laplace Equation for Pressure

Theorem 2.1 (Pressure Laplacian). *For an incompressible fluid, the pressure satisfies Laplace's equation outside the source:*

$$\nabla^2 p = 0 \quad (2.5)$$

Proof. For incompressible potential flow with $\vec{v} = -\nabla\phi$:

$$\nabla \cdot \vec{v} = 0 \quad (2.6)$$

$$\nabla \cdot (-\nabla\phi) = 0 \quad (2.7)$$

$$-\nabla^2\phi = 0 \quad (2.8)$$

$$\nabla^2\phi = 0 \quad (2.9)$$

For a barotropic fluid where $p = p(\rho)$ with constant ρ , the pressure satisfies the same equation:

$$\nabla^2 p = 0 \quad (\text{outside source}) \quad (2.10)$$

□

Dimensional Check

$$[\nabla^2 p] = \frac{[p]}{[L]^2} = \frac{\text{Pa}}{\text{m}^2} = \frac{\text{kg}}{\text{m}^3 \cdot \text{s}^2} \quad \checkmark \quad (2.11)$$

2.3 General Solution in Spherical Coordinates

Theorem 2.2 (Spherical Solution). *The general spherically symmetric solution to $\nabla^2 p = 0$ is:*

$$p(r) = A + \frac{B}{r} \quad (2.12)$$

where A and B are constants determined by boundary conditions.

Proof. In spherical coordinates, for a function $f(r)$ depending only on r :

$$\nabla^2 f = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{df}{dr} \right) \quad (2.13)$$

Expanding:

$$\nabla^2 f = \frac{1}{r^2} \left[2r \frac{df}{dr} + r^2 \frac{d^2 f}{dr^2} \right] = \frac{2}{r} \frac{df}{dr} + \frac{d^2 f}{dr^2} \quad (2.14)$$

For $\nabla^2 p = 0$:

$$\frac{d^2 p}{dr^2} + \frac{2}{r} \frac{dp}{dr} = 0 \quad (2.15)$$

This is a second-order ODE. Substituting $q = dp/dr$:

$$\frac{dq}{dr} + \frac{2}{r} q = 0 \quad (2.16)$$

$$\frac{dq}{q} = -\frac{2}{r} \frac{dr}{r} \quad (2.17)$$

$$\ln |q| = -2 \ln |r| + C_1 \quad (2.18)$$

$$q = \frac{A'}{r^2} \quad (2.19)$$

Integrating:

$$p = \int \frac{A'}{r^2} dr = -\frac{A'}{r} + B = A + \frac{B'}{r} \quad (2.20)$$

(Relabeling constants for convenience.) □

Verification by substitution:

$$\frac{dp}{dr} = -\frac{B'}{r^2} \quad (2.21)$$

$$\frac{d^2 p}{dr^2} = \frac{2B'}{r^3} \quad (2.22)$$

$$\frac{d^2 p}{dr^2} + \frac{2}{r} \frac{dp}{dr} = \frac{2B'}{r^3} + \frac{2}{r} \left(-\frac{B'}{r^2} \right) = \frac{2B'}{r^3} - \frac{2B'}{r^3} = 0 \quad \checkmark \quad (2.23)$$

2.4 Boundary Conditions

2.4.1 Condition at Infinity

At large distances from the source, the pressure approaches the background Plenum pressure:

$$p(r \rightarrow \infty) = p_\infty \quad (2.24)$$

From eq. (2.12):

$$p_\infty = A + \frac{B'}{r} = A \implies A = p_\infty \quad (2.25)$$

2.4.2 Condition at Core

At the vortex core, pressure is completely excluded (Postulate 2.4):

$$p(r_{\text{core}}) = 0 \quad (2.26)$$

Substituting:

$$0 = p_\infty + \frac{B'}{r_{\text{core}}} \implies B' = -p_\infty \cdot r_{\text{core}} \quad (2.27)$$

2.4.3 Complete Pressure Profile

Pressure Profile

$$p(r) = p_\infty \left(1 - \frac{r_{\text{core}}}{r} \right) \quad (2.28)$$

Dimensional Check

$$[p] = [p_\infty] \left[1 - \frac{r_{\text{core}}}{r} \right] = \text{Pa} \times [1] = \text{Pa} \quad \checkmark \quad (2.29)$$

2.5 Euler Equation for Velocity

Theorem 2.3 (Steady Euler Equation). *For stationary, inviscid flow, the Euler equation in vector form is:*

$$\rho(\vec{v} \cdot \nabla)\vec{v} = -\nabla p \quad (2.30)$$

For purely radial flow $\vec{v} = v(r)\hat{r}$:

$$(\vec{v} \cdot \nabla)\vec{v} = v \frac{dv}{dr} \hat{r} \quad (2.31)$$

The radial component of the Euler equation becomes:

$$\rho v \frac{dv}{dr} = -\frac{dp}{dr} \quad (2.32)$$

Dimensional Check

$$\text{LHS: } \left[\rho v \frac{dv}{dr} \right] = \frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}}{\text{s}} \cdot \frac{1}{\text{s}} = \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \quad (2.33)$$

$$\text{RHS: } \left[\frac{dp}{dr} \right] = \frac{\text{Pa}}{\text{m}} = \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \quad \checkmark \quad (2.34)$$

2.6 Pressure Gradient

From eq. (2.28):

$$\frac{dp}{dr} = p_\infty \cdot \frac{r_{\text{core}}}{r^2} \quad (2.35)$$

Dimensional Check

$$\left[\frac{dp}{dr} \right] = \frac{\text{Pa} \cdot \text{m}}{\text{m}^2} = \frac{\text{Pa}}{\text{m}} \quad \checkmark \quad (2.36)$$

2.7 Integration of Euler Equation

Substituting eq. (2.35) into eq. (2.32):

$$\rho v \frac{dv}{dr} = - \frac{p_\infty \cdot r_{\text{core}}}{r^2} \quad (2.37)$$

Rearranging:

$$v \, dv = - \frac{p_\infty}{\rho} \cdot \frac{r_{\text{core}}}{r^2} \, dr \quad (2.38)$$

Integrating both sides:

$$\int v \, dv = - \frac{p_\infty}{\rho} r_{\text{core}} \int \frac{dr}{r^2} \quad (2.39)$$

$$\frac{v^2}{2} = \frac{p_\infty}{\rho} \cdot \frac{r_{\text{core}}}{r} + C_2 \quad (2.40)$$

2.7.1 Boundary Condition for Velocity

At infinity, the flow velocity vanishes:

$$v(r \rightarrow \infty) = 0 \quad (2.41)$$

Therefore:

$$0 = \frac{p_\infty}{\rho} \cdot \frac{r_{\text{core}}}{\infty} + C_2 = 0 + C_2 \implies C_2 = 0 \quad (2.42)$$

2.7.2 Velocity Profile

$$v^2 = \frac{2p_\infty}{\rho} \cdot \frac{r_{\text{core}}}{r} \quad (2.43)$$

2.8 Equation of State

Postulate 2.5 (Stiff Equation of State). The Plenum obeys a relativistic (stiff) equation of state:

$$p_{\infty} = \rho c^2 \quad (2.44)$$

Dimensional Check

$$[\rho c^2] = \frac{\text{kg}}{\text{m}^3} \cdot \frac{\text{m}^2}{\text{s}^2} = \frac{\text{kg}}{\text{m} \cdot \text{s}^2} = \text{Pa} \quad \checkmark \quad (2.45)$$

Substituting eq. (2.44) into eq. (2.43):

$$v^2 = \frac{2\rho c^2}{\rho} \cdot \frac{r_{\text{core}}}{r} = 2c^2 \cdot \frac{r_{\text{core}}}{r} \quad (2.46)$$

$$v = c \sqrt{\frac{2r_{\text{core}}}{r}} \quad (2.47)$$

2.9 Identification of Core Radius

Comparing with the Newtonian free-fall velocity:

$$v_{\text{Newton}} = \sqrt{\frac{2GM}{r}} \quad (2.48)$$

Equating eq. (2.47) and eq. (2.48):

$$c \sqrt{\frac{2r_{\text{core}}}{r}} = \sqrt{\frac{2GM}{r}} \quad (2.49)$$

$$c^2 \cdot \frac{2r_{\text{core}}}{r} = \frac{2GM}{r} \quad (2.50)$$

$$c^2 r_{\text{core}} = GM \quad (2.51)$$

Core Radius Identification

$$r_{\text{core}} = \frac{GM}{c^2} = \frac{r_s}{2} \quad (2.52)$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius.

Dimensional Check

$$\left[\frac{GM}{c^2}\right] = \frac{\text{m}^3/(\text{kg} \cdot \text{s}^2) \cdot \text{kg}}{\text{m}^2/\text{s}^2} = \text{m} \quad \checkmark \quad (2.53)$$

2.10 Main Result: Plenum Flow Velocity

Plenum Flow Velocity

The velocity of Plenum flow toward a gravitational source of mass M is:

$$v(r) = \sqrt{\frac{2GM}{r}} \quad (2.54)$$

Physical interpretation: Objects on the membrane “float” with this flow, experiencing what we perceive as gravitational attraction.

Remark 2.1. This velocity profile is identical to the Newtonian free-fall velocity and to the Painlevé-Gullstrand form of the Schwarzschild metric. EDC thus naturally connects to General Relativity in the weak-field limit.

2.11 Summary of Derivation Steps

1. Started with incompressibility $\nabla \cdot \vec{v} = 0$
2. Derived Laplace equation $\nabla^2 p = 0$
3. Solved in spherical coordinates: $p(r) = A + B/r$
4. Applied boundary conditions: $p(\infty) = p_\infty$, $p(r_{\text{core}}) = 0$
5. Obtained pressure profile: $p(r) = p_\infty(1 - r_{\text{core}}/r)$
6. Applied Euler equation: $\rho v dv/dr = -dp/dr$
7. Integrated with $v(\infty) = 0$
8. Used equation of state $p_\infty = \rho c^2$
9. Identified $r_{\text{core}} = GM/c^2$ by comparison with Newton

Epistemic status: **[D]** — Mathematically derived from explicit assumptions.

Chapter 3

Superposition of Gravitational Sources

3.1 Statement of the Theorem

Theorem 3.1 (Superposition of Masses). *For N vortices (particles) with masses $M_1, M_2, \dots, M_N$ located at positions $\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N$, the flow velocity in the far field ($r \gg \max |\vec{r}_i - \vec{r}_j|$) is:*

$$v(r) = \sqrt{\frac{2G \cdot M_{total}}{r}} \quad (3.1)$$

where

$$M_{total} = \sum_{i=1}^N M_i \quad (3.2)$$

3.2 Proof

3.2.1 Step 1: Linearity of the Laplacian

Lemma 3.2 (Linearity). *The Laplacian operator ∇^2 is linear:*

$$\nabla^2(\alpha f + \beta g) = \alpha \nabla^2 f + \beta \nabla^2 g \quad (3.3)$$

Proof.

$$\nabla^2(\alpha f + \beta g) = \frac{\partial^2(\alpha f + \beta g)}{\partial x^2} + \frac{\partial^2(\alpha f + \beta g)}{\partial y^2} + \frac{\partial^2(\alpha f + \beta g)}{\partial z^2} \quad (3.4)$$

$$= \alpha \left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} \right) + \beta \left(\frac{\partial^2 g}{\partial x^2} + \frac{\partial^2 g}{\partial y^2} + \frac{\partial^2 g}{\partial z^2} \right) \quad (3.5)$$

$$= \alpha \nabla^2 f + \beta \nabla^2 g \quad (3.6)$$

□

3.2.2 Step 2: Superposition of Pressures

Let $p_i(\vec{r})$ be the pressure field due to source i alone, satisfying $\nabla^2 p_i = 0$.

Define the total pressure:

$$p_{total}(\vec{r}) = \sum_{i=1}^N p_i(\vec{r}) \quad (3.7)$$

Then by linearity:

$$\nabla^2 p_{\text{total}} = \nabla^2 \left(\sum_{i=1}^N p_i \right) = \sum_{i=1}^N \nabla^2 p_i = \sum_{i=1}^N 0 = 0 \quad (3.8)$$

3.2.3 Step 3: Individual Source Contributions

For source i at position $\vec{r}_i$ with mass M_i :

$$p_i(\vec{r}) = p_\infty \left(1 - \frac{r_{\text{core},i}}{|\vec{r} - \vec{r}_i|} \right) \quad (3.9)$$

where

$$r_{\text{core},i} = \frac{GM_i}{c^2} \quad (3.10)$$

The pressure deficit relative to p_∞ :

$$\Delta p_i(\vec{r}) = p_i(\vec{r}) - p_\infty = -\frac{p_\infty \cdot r_{\text{core},i}}{|\vec{r} - \vec{r}_i|} \quad (3.11)$$

3.2.4 Step 4: Total Pressure

$$p_{\text{total}}(\vec{r}) = p_\infty + \sum_{i=1}^N \Delta p_i(\vec{r}) = p_\infty \left(1 - \sum_{i=1}^N \frac{r_{\text{core},i}}{|\vec{r} - \vec{r}_i|} \right) \quad (3.12)$$

3.2.5 Step 5: Far-Field (Multipole) Expansion

For $|\vec{r}| \gg |\vec{r}_i|$, Taylor expand:

$$\frac{1}{|\vec{r} - \vec{r}_i|} = \frac{1}{\sqrt{(\vec{r} - \vec{r}_i) \cdot (\vec{r} - \vec{r}_i)}} \approx \frac{1}{r} + \frac{\hat{r} \cdot \vec{r}_i}{r^2} + O\left(\frac{|\vec{r}_i|^2}{r^3}\right) \quad (3.13)$$

Derivation:

$$|\vec{r} - \vec{r}_i|^2 = r^2 - 2\vec{r} \cdot \vec{r}_i + r_i^2 = r^2 \left(1 - \frac{2\hat{r} \cdot \vec{r}_i}{r} + \frac{r_i^2}{r^2} \right) \quad (3.14)$$

Let $\varepsilon = -2(\hat{r} \cdot \vec{r}_i)/r + r_i^2/r^2$ with $|\varepsilon| \ll 1$:

$$\frac{1}{|\vec{r} - \vec{r}_i|} = \frac{1}{r} (1 + \varepsilon)^{-1/2} \quad (3.15)$$

$$\approx \frac{1}{r} \left(1 - \frac{\varepsilon}{2} + O(\varepsilon^2) \right) \quad (3.16)$$

$$= \frac{1}{r} \left(1 + \frac{\hat{r} \cdot \vec{r}_i}{r} + O(1/r^2) \right) \quad (3.17)$$

3.2.6 Step 6: Monopole Dominance

Retaining only the $1/r$ term:

$$\sum_{i=1}^N \frac{r_{\text{core},i}}{|\vec{r} - \vec{r}_i|} \approx \sum_{i=1}^N \frac{r_{\text{core},i}}{r} = \frac{1}{r} \sum_{i=1}^N r_{\text{core},i} \quad (3.18)$$

Define the total core radius:

$$r_{\text{core,total}} = \sum_{i=1}^N r_{\text{core},i} = \sum_{i=1}^N \frac{GM_i}{c^2} = \frac{G}{c^2} \sum_{i=1}^N M_i = \frac{G \cdot M_{\text{total}}}{c^2} \quad (3.19)$$

3.2.7 Step 7: Far-Field Pressure

$$p_{\text{total}}(r) \approx p_{\infty} \left(1 - \frac{r_{\text{core,total}}}{r} \right) \quad (3.20)$$

This has **exactly the same form** as the single-source case with $M = M_{\text{total}}$.

3.2.8 Step 8: Velocity from Euler Equation

Following the same procedure as Chapter 2:

$$v(r) = \sqrt{\frac{2G \cdot M_{\text{total}}}{r}} \quad (3.21)$$

Superposition Theorem

$$M_{\text{total}} = \sum_{i=1}^N M_i \quad (3.22)$$

Newtonian superposition is a **consequence** of the linearity of Laplace's equation in EDC.

3.3 Regime of Validity

Condition	Criterion	Error if violated
Far field	$r > 10 \times \max \vec{r}_i - \vec{r}_j $	$< 1\%$
Weak field	$r > 10 \times r_{s,\text{total}}$	Relativistic corrections
Non-overlapping cores	$ \vec{r}_i - \vec{r}_j \gg r_{\text{core},i} + r_{\text{core},j}$	Topological interference

Table 3.1: Validity conditions for the superposition theorem.

3.4 Comparison with General Relativity

In General Relativity, the Einstein field equations are **nonlinear**:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (3.23)$$

Superposition is only approximate in GR, valid in the weak-field limit.

In EDC, superposition of pressures is **exact** (for the pressure field), because Laplace's equation is linear. The far-field velocity inherits this superposition property.

Epistemic status: **[D]** — Mathematically derived from linearity of ∇^2 .

Chapter 4

Upper Bound on Plenum Viscosity

The derivations in Chapters 2 and 3 assumed inviscid flow. This chapter justifies that assumption by deriving an upper bound on the kinematic viscosity of the Plenum.

4.1 Navier-Stokes with Viscosity

For steady viscous flow:

$$\rho(\vec{v} \cdot \nabla)\vec{v} = -\nabla p + \eta \nabla^2 \vec{v} \quad (4.1)$$

Dimensional Check

$$[\eta \nabla^2 v] = \frac{\text{Pa} \cdot \text{s} \cdot \text{m/s}}{\text{m}^2} = \frac{\text{kg}}{\text{m}^2 \cdot \text{s}^2} \quad \checkmark \quad (4.2)$$

4.2 Vector Laplacian in Spherical Coordinates

For radial flow $\vec{v} = v(r)\hat{r}$:

$$\nabla^2 \vec{v} = \left[\frac{d^2 v}{dr^2} + \frac{2}{r} \frac{dv}{dr} - \frac{2v}{r^2} \right] \hat{r} \quad (4.3)$$

4.3 Derivatives of the Inviscid Solution

For $v_0(r) = \sqrt{2GM/r} = \sqrt{2GM} \cdot r^{-1/2}$:

$$\frac{dv_0}{dr} = \sqrt{2GM} \cdot \left(-\frac{1}{2} \right) r^{-3/2} = -\frac{v_0}{2r} \quad (4.4)$$

$$\frac{d^2 v_0}{dr^2} = \sqrt{2GM} \cdot \frac{3}{4} r^{-5/2} = \frac{3v_0}{4r^2} \quad (4.5)$$

Dimensional Check

$$\left[\frac{dv}{dr} \right] = \frac{\text{m/s}}{\text{m}} = \text{s}^{-1} \quad \checkmark \quad \left[\frac{d^2 v}{dr^2} \right] = \frac{\text{m/s}}{\text{m}^2} = (\text{m} \cdot \text{s})^{-1} \quad \checkmark \quad (4.6)$$

4.4 Laplacian of Inviscid Solution

$$\nabla^2 v_0 = \frac{3v_0}{4r^2} + \frac{2}{r} \left(-\frac{v_0}{2r} \right) - \frac{2v_0}{r^2} \quad (4.7)$$

$$= \frac{v_0}{r^2} \left(\frac{3}{4} - 1 - 2 \right) \quad (4.8)$$

$$= -\frac{9v_0}{4r^2} \quad (4.9)$$

4.5 Perturbative Analysis

Ansatz:

$$v = v_0 + \delta v \quad \text{where } |\delta v| \ll |v_0| \quad (4.10)$$

Substituting into Navier-Stokes and keeping first-order terms:

Zeroth order (inviscid solution):

$$\rho v_0 \frac{dv_0}{dr} = -\frac{dp}{dr} \quad (4.11)$$

First order (viscous correction):

$$\rho v_0 \frac{d\delta v}{dr} + \rho \delta v \frac{dv_0}{dr} = \eta \nabla^2 v_0 \quad (4.12)$$

Substituting eq. (4.4) and eq. (4.9):

$$\rho v_0 \frac{d\delta v}{dr} + \rho \delta v \left(-\frac{v_0}{2r} \right) = \eta \left(-\frac{9v_0}{4r^2} \right) \quad (4.13)$$

Dividing by ρv_0 :

$$\frac{d\delta v}{dr} - \frac{\delta v}{2r} = -\frac{9\eta}{4\rho r^2} \quad (4.14)$$

4.6 Solution of the ODE

Equation eq. (4.14) is a linear first-order ODE: $\frac{d\delta v}{dr} + P(r)\delta v = Q(r)$

With $P(r) = -1/(2r)$ and $Q(r) = -9\eta/(4\rho r^2)$.

Integrating factor:

$$\mu(r) = \exp \left(\int P dr \right) = \exp \left(-\int \frac{dr}{2r} \right) = \exp \left(-\frac{1}{2} \ln r \right) = r^{-1/2} \quad (4.15)$$

Multiplying by μ :

$$\frac{d}{dr} [r^{-1/2} \delta v] = -\frac{9\eta}{4\rho r^{5/2}} \quad (4.16)$$

Integrating:

$$r^{-1/2} \delta v = -\frac{9\eta}{4\rho} \int r^{-5/2} dr = -\frac{9\eta}{4\rho} \cdot \left(-\frac{2}{3} \right) r^{-3/2} + C \quad (4.17)$$

$$r^{-1/2}\delta v = \frac{3\eta}{2\rho} \cdot r^{-3/2} + C \quad (4.18)$$

Multiplying by $r^{1/2}$:

$$\delta v = \frac{3\eta}{2\rho r} + C \cdot r^{1/2} \quad (4.19)$$

4.6.1 Boundary Condition

For $\delta v(r \rightarrow \infty) = 0$:

$$C = 0 \quad (\text{otherwise } \delta v \text{ diverges}) \quad (4.20)$$

Viscous Correction

$$\delta v(r) = \frac{3\eta}{2\rho r} = \frac{3\nu_{\text{bulk}}}{2r} \quad (4.21)$$

where $\nu_{\text{bulk}} = \eta/\rho$ is the kinematic viscosity.

Dimensional Check

$$\left[\frac{\eta}{\rho r}\right] = \frac{\text{Pa} \cdot \text{s}}{\text{kg}/\text{m}^3 \cdot \text{m}} = \frac{\text{kg}/(\text{m} \cdot \text{s})}{\text{kg}/\text{m}^2} = \text{m}/\text{s} \quad \checkmark \quad (4.22)$$

4.7 Relative Correction

$$\frac{\delta v}{v_0} = \frac{3\nu_{\text{bulk}}/(2r)}{\sqrt{2GM/r}} = \frac{3\nu_{\text{bulk}}}{2r} \cdot \sqrt{\frac{r}{2GM}} = \frac{3\nu_{\text{bulk}}}{2\sqrt{2GM}r} \quad (4.23)$$

4.8 Application to Mercury's Orbit

Constraint: The viscous correction must be smaller than the observational precision of Mercury's perihelion precession ($\sim 0.022\%$).

Conservatively require: $|\delta v/v_0| < \varepsilon = 10^{-4}$

For Mercury:

$$a_{\text{Mercury}} = 5.79 \times 10^{10} \text{ m} \quad (4.24)$$

$$M_{\odot} = 1.989 \times 10^{30} \text{ kg} \quad (4.25)$$

$$GM_{\odot} = 1.327 \times 10^{20} \text{ m}^3/\text{s}^2 \quad (4.26)$$

From eq. (4.23):

$$\frac{3\nu_{\text{bulk}}}{2\sqrt{2GM_{\odot}} \cdot a} < 10^{-4} \quad (4.27)$$

$$\nu_{\text{bulk}} < \frac{10^{-4} \cdot 2\sqrt{2 \times 1.327 \times 10^{20} \times 5.79 \times 10^{10}}}{3} \quad (4.28)$$

$$\nu_{\text{bulk}} < \frac{10^{-4} \cdot 2\sqrt{1.537 \times 10^{31}}}{3} = \frac{10^{-4} \cdot 2 \cdot 3.92 \times 10^{15}}{3} \quad (4.29)$$

Viscosity Upper Bound

$$\nu_{\text{bulk}} \leq 2.6 \times 10^{11} \text{ m}^2/\text{s} \quad (4.30)$$

4.9 Reynolds Number

The Reynolds number at Mercury's orbit:

$$\text{Re} = \frac{v_0 \cdot L}{\nu_{\text{bulk}}} \quad (4.31)$$

With $v_0 \sim 50 \text{ km/s}$, $L \sim a \sim 6 \times 10^{10} \text{ m}$:

$$\text{Re} > \frac{5 \times 10^4 \times 6 \times 10^{10}}{2.6 \times 10^{11}} \sim 10^4 \quad (4.32)$$

Inviscid Approximation Justified

$\text{Re} \gg 1$ confirms that the Plenum flow is **effectively inviscid** in the Solar System regime.

Epistemic status: **[D]** — Derived from perturbative analysis and Mercury constraint.

Chapter 5

Connection to Quantum Mechanics

This chapter presents the identified relationships between EDC parameters and fundamental quantum constants.

5.1 Planck Constant from Membrane Tension

Planck Constant

$$\hbar = \frac{\sigma r_e^3}{c} \quad (5.1)$$

5.1.1 Dimensional Check

Dimensional Check

$$[\sigma r_e^3 / c] = [MT^{-2}][L^3][L^{-1}T] \quad (5.2)$$

$$= [ML^2T^{-1}] \quad (5.3)$$

$$= [\hbar] \quad \checkmark \quad (5.4)$$

5.1.2 Numerical Verification

$$\sigma = 1.41 \times 10^{18} \text{ kg/s}^2 \quad (5.5)$$

$$r_e = 2.8179 \times 10^{-15} \text{ m} \quad (5.6)$$

$$r_e^3 = 2.237 \times 10^{-44} \text{ m}^3 \quad (5.7)$$

$$c = 2.998 \times 10^8 \text{ m/s} \quad (5.8)$$

$$\hbar_{\text{EDC}} = \frac{1.41 \times 10^{18} \times 2.237 \times 10^{-44}}{2.998 \times 10^8} = 1.052 \times 10^{-34} \text{ J} \cdot \text{s} \quad (5.9)$$

$$\hbar_{\text{CODATA}} = 1.055 \times 10^{-34} \text{ J} \cdot \text{s} \quad (5.10)$$

Match: 99.7%

Note 5.1. The membrane tension $\sigma = 1.41 \times 10^{18} \text{ J/m}^2$ is **calibrated** to make this relationship exact. This is why the epistemic status is [I] rather than [D].

5.2 Fine Structure Constant

Fine Structure Constant

$$\alpha = \frac{m_e c^2}{\sigma r_e^2} \quad (5.11)$$

5.2.1 Dimensional Check

Dimensional Check

$$\left[\frac{m_e c^2}{\sigma r_e^2} \right] = \frac{[M][L^2 T^{-2}]}{[M T^{-2}][L^2]} \quad (5.12)$$

$$= \frac{[M L^2 T^{-2}]}{[M L^2 T^{-2}]} \quad (5.13)$$

$$= [1] \quad \checkmark \text{ (dimensionless)} \quad (5.14)$$

5.2.2 Numerical Verification

$$m_e c^2 = 9.109 \times 10^{-31} \times (2.998 \times 10^8)^2 = 8.187 \times 10^{-14} \text{ J} \quad (5.15)$$

$$\sigma r_e^2 = 1.41 \times 10^{18} \times 7.94 \times 10^{-30} = 1.120 \times 10^{-11} \text{ J} \quad (5.16)$$

$$\alpha_{\text{EDC}} = \frac{8.187 \times 10^{-14}}{1.120 \times 10^{-11}} = 7.31 \times 10^{-3} = \frac{1}{136.8} \quad (5.17)$$

$$\alpha_{\text{CODATA}} = \frac{1}{137.036} = 7.297 \times 10^{-3} \quad (5.18)$$

Match: 99.8%

5.3 Vortex Elastic Energy and Electron Mass

5.3.1 Definition of Vortex Energy

Definition 5.1 (Vortex Elastic Energy). The elastic energy associated with a vortex on the membrane is:

$$\mathcal{E}_v = \sigma r_e^2 \quad (5.19)$$

Dimensional Check

$$[\sigma r_e^2] = [M T^{-2}][L^2] = [M L^2 T^{-2}] = \text{J} \quad \checkmark \quad (5.20)$$

5.3.2 Equivalent Mass

Definition 5.2 (Vortex Equivalent Mass). The equivalent mass of the vortex elastic energy (via $E = mc^2$):

$$\mathcal{M}_v = \frac{\mathcal{E}_v}{c^2} = \frac{\sigma r_e^2}{c^2} \quad (5.21)$$

5.3.3 Derivation of Electron Mass Relation

Theorem 5.1 (Electron Mass from Vortex Energy).

$$m_e = \alpha \cdot \mathcal{M}_v = \frac{\alpha \sigma r_e^2}{c^2} \quad (5.22)$$

Proof. From the standard definition of the classical electron radius:

$$r_e = \frac{\alpha \hbar}{m_e c} \quad (5.23)$$

From the EDC identification:

$$\hbar = \frac{\sigma r_e^3}{c} \quad (5.24)$$

Substituting eq. (5.24) into eq. (5.23):

$$r_e = \frac{\alpha \cdot (\sigma r_e^3 / c)}{m_e c} = \frac{\alpha \sigma r_e^3}{m_e c^2} \quad (5.25)$$

Solving for m_e :

$$m_e c^2 = \frac{\alpha \sigma r_e^3}{r_e} = \alpha \sigma r_e^2 \quad (5.26)$$

$$m_e = \frac{\alpha \sigma r_e^2}{c^2} = \alpha \cdot \mathcal{M}_v \quad (5.27)$$

□

Electron Mass

$$m_e = \alpha \cdot \mathcal{M}_v = \frac{\alpha \sigma r_e^2}{c^2} \quad (5.28)$$

The electron mass is exactly α times the equivalent mass of the vortex elastic energy.

5.3.4 The Ratio $\mathcal{M}_v/m_e$

$$\frac{\mathcal{M}_v}{m_e} = \frac{\sigma r_e^2 / c^2}{\alpha \sigma r_e^2 / c^2} = \frac{1}{\alpha} = 137.036 \quad (5.29)$$

Energy Ratio

$$\boxed{\frac{\mathcal{M}_v}{m_e} = \frac{1}{\alpha} \approx 137} \quad (5.30)$$

Physical interpretation: The total elastic energy of a vortex ($\mathcal{E}_v = \sigma r_e^2 \approx 70$ MeV) is 137 times larger than the rest mass energy of the electron ($m_e c^2 = 0.511$ MeV).

Only a fraction $\alpha \approx 1/137$ of the vortex elastic energy manifests as inertial mass.

5.3.5 Numerical Values

$$\mathcal{E}_v = \sigma r_e^2 = 1.12 \times 10^{-11} \text{ J} = 70.1 \text{ MeV} \quad (5.31)$$

$$\mathcal{M}_v = \mathcal{E}_v / c^2 = 1.25 \times 10^{-28} \text{ kg} \quad (5.32)$$

$$m_e = 9.11 \times 10^{-31} \text{ kg} = 0.511 \text{ MeV} / c^2 \quad (5.33)$$

$$\frac{\mathcal{M}_v}{m_e} = 136.9 \approx \frac{1}{\alpha} \quad (5.34)$$

Epistemic status: The relation $m_e = \alpha \mathcal{M}_v$ is **[D]** from the combination of $\hbar = \sigma r_e^3 / c$ and $r_e = \alpha \hbar / (m_e c)$. The ratio $\mathcal{M}_v / m_e = 1/\alpha$ is a mathematical consequence, not an independent discovery.

Chapter 6

Newton's Gravitational Constant

This chapter presents the identified formula for Newton's gravitational constant G in terms of EDC parameters.

6.1 The Formula

Gravitational Constant

$$G = \frac{c^4 R_\xi^{12}}{128\pi^2 \sigma r_e^{13}} \quad (6.1)$$

Critical Note on Epistemic Status

The powers 12 and 13, and the factor $128\pi^2$, were found by **numerical fitting**, not derived from first principles. This formula has status **[I]**, not **[D]**.

6.2 Dimensional Analysis

Dimensional Check

$$[G] = \frac{[c^4][R_\xi^{12}]}{[\sigma][r_e^{13}]} \quad (6.2)$$

$$= \frac{[L^4 T^{-4}][L^{12}]}{[M T^{-2}][L^{13}]} \quad (6.3)$$

$$= \frac{[L^{16} T^{-4}]}{[M L^{13} T^{-2}]} \quad (6.4)$$

$$= [L^3 M^{-1} T^{-2}] \quad \checkmark \quad (6.5)$$

6.3 Numerical Verification

$$c^4 = (2.998 \times 10^8)^4 = 8.077 \times 10^{33} \text{ m}^4/\text{s}^4 \quad (6.6)$$

$$R_\xi^{12} = (2.16 \times 10^{-18})^{12} = 1.19 \times 10^{-213} \text{ m}^{12} \quad (6.7)$$

$$128\pi^2 = 1263.3 \quad (6.8)$$

$$\sigma = 1.41 \times 10^{18} \text{ kg/s}^2 \quad (6.9)$$

$$r_e^{13} = (2.8179 \times 10^{-15})^{13} = 3.98 \times 10^{-188} \text{ m}^{13} \quad (6.10)$$

$$\text{Numerator} = 8.077 \times 10^{33} \times 1.19 \times 10^{-213} = 9.61 \times 10^{-180} \quad (6.11)$$

$$\text{Denominator} = 1263.3 \times 1.41 \times 10^{18} \times 3.98 \times 10^{-188} = 7.09 \times 10^{-167} \quad (6.12)$$

$$G_{\text{EDC}} = \frac{9.61 \times 10^{-180}}{7.09 \times 10^{-167}} = 6.62 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) \quad (6.13)$$

$$G_{\text{CODATA}} = 6.674 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2) \quad (6.14)$$

Error: 0.81%

6.4 Sensitivity Analysis

The 0.81% error can be eliminated by adjusting R_ξ by only 0.07%:

$$R_{\xi \text{ exact}} = 2.1615 \times 10^{-18} \text{ m} \quad (6.15)$$

This is well within the uncertainty of the EDC parameter.

6.5 The Hierarchy Factor

The factor $(R_\xi/r_e)^{12}$ encodes the enormous hierarchy between gravity and electromagnetism:

$$\frac{R_\xi}{r_e} = \frac{2.16 \times 10^{-18}}{2.82 \times 10^{-15}} = 7.66 \times 10^{-4} \quad (6.16)$$

$$\left(\frac{R_\xi}{r_e}\right)^{12} = (7.66 \times 10^{-4})^{12} = 4.1 \times 10^{-38} \quad (6.17)$$

Compare to the gravitational fine structure constant:

$$\alpha_G = \frac{Gm_e^2}{\hbar c} = 1.75 \times 10^{-45} \quad (6.18)$$

The ratio $(R_\xi/r_e)^{12} \approx 10^{-38}$ is close to the gravity/EM hierarchy.

6.6 Proposed Interpretations

Interpretation of Power 12

$$12 = 4 \times 3 \quad (\text{spacetime dimensions} \times \text{spatial dimensions}) \quad (6.19)$$

In 5D EDC geometry, the effective gravitational coupling may involve integration over 4D spacetime with additional factors from 3D spatial structure.

Interpretation of Power 13

$$13 = 12 + 1 \quad (+ \text{compact dimension contribution}) \quad (6.20)$$

The extra power may arise from normalization over the compact ξ dimension.

Interpretation of $128\pi^2$

$$128\pi^2 = (4\pi)^2 \times 8 = 16\pi^2 \times 8 \quad (6.21)$$

- $(4\pi)^2$: Double application of 3D Gauss's law
- $8 = 2^3$: Factor of 2 for each spatial dimension

Status of Interpretations

All interpretations in this section have status **[P]**. They are physically motivated but **not rigorously derived** from the 5D EDC action. A rigorous derivation of the powers 12, 13 and the factor $128\pi^2$ remains an **open problem**.

Chapter 7

The Two Radii of a Particle

One of the key insights from EDC is that particles have two distinct characteristic radii.

7.1 Topological Radius

Definition 7.1 (Topological Radius). The topological radius r_{topo} is the physical size of the vortex on the membrane:

$$r_{\text{topo}} \sim r_e \sim 10^{-15} \text{ m} \quad (\text{for the electron}) \quad (7.1)$$

This radius determines:

- Electromagnetic interactions
- Quantum mechanical properties ($\hbar = \sigma r_e^3 / c$)
- The fine structure constant ($\alpha = m_e c^2 / (\sigma r_e^2)$)

7.2 Gravitational Radius

Definition 7.2 (Gravitational Radius). The gravitational radius r_{grav} is the size of the pressure deficit in the Plenum:

$$r_{\text{grav}} = \frac{Gm}{c^2} \sim 10^{-58} \text{ m} \quad (\text{for the electron}) \quad (7.2)$$

This radius determines:

- The gravitational field strength
- The core radius r_{core} where $p = 0$

7.3 The Hierarchy as a Ratio

Hierarchy from Two Radii

$$\frac{r_{\text{topo}}}{r_{\text{grav}}} = \frac{r_e}{Gm_e/c^2} = \frac{r_e c^2}{Gm_e} \quad (7.3)$$

Numerically:

$$\frac{r_{\text{topo}}}{r_{\text{grav}}} = \frac{2.82 \times 10^{-15}}{6.76 \times 10^{-58}} = 4.17 \times 10^{42} \quad (7.4)$$

This is exactly the gravity/EM hierarchy!

Physical Interpretation

Gravity is negligible at atomic scales because the gravitational radius is 10^{42} times smaller than the topological radius.

The hierarchy is not “fine-tuning” — it is a **geometric fact** in EDC.

7.4 Summary: Three Scales of the Electron

Scale	Formula	Value	Physical meaning
Classical radius	$r_e = \alpha \hbar / (m_e c)$	$2.82 \times 10^{-15} \text{ m}$	EM/topological size
Compton wavelength	$\lambda_C = \hbar / (m_e c)$	$3.86 \times 10^{-13} \text{ m}$	Quantum localization
Schwarzschild radius	$r_s = 2Gm_e/c^2$	$1.35 \times 10^{-57} \text{ m}$	Gravitational size

Table 7.1: The three characteristic scales of the electron.

Relations:

$$\frac{r_e}{\lambda_C} = \alpha = \frac{1}{137} \quad (\text{fine structure}) \quad (7.5)$$

$$\frac{r_e}{r_s} = 2.1 \times 10^{42} \quad (\text{hierarchy!}) \quad (7.6)$$

Chapter 8

Summary and Epistemic Status

8.1 What Has Been Derived (Status D)

Result	Equation	Chapter
Plenum flow velocity	$v(r) = \sqrt{2GM/r}$	2
Superposition of masses	$M_{\text{total}} = \sum M_i$	3
Viscosity upper bound	$\nu_{\text{bulk}} \leq 2.6 \times 10^{11} \text{ m}^2/\text{s}$	4
Electron mass relation	$m_e = \alpha \cdot \mathcal{M}_v$	5
Energy ratio	$\mathcal{M}_v/m_e = 1/\alpha$	5

Table 8.1: Results with status **[D]**.

8.2 What Has Been Identified (Status I)

Result	Formula	Match	Note
Planck constant	$\hbar = \sigma r_e^3/c$	99.7%	σ calibrated
Fine structure	$\alpha = m_e c^2 / (\sigma r_e^2)$	99.8%	Consistent
Newton's constant	$G = c^4 R_\xi^{12} / (128 \pi^2 \sigma r_e^{13})$	99.2%	Powers fitted

Table 8.2: Results with status **[I]**.

8.3 What Has Been Proposed (Status P)

Proposal	Status
Power 12 = 4×3 (spacetime $\times$ space)	Speculation
Power 13 = $12 + 1$ (+ compact dimension)	Speculation
$128\pi^2 = (4\pi)^2 \times 8$	Post hoc interpretation
Particles as vortices	Foundational postulate

Table 8.3: Results with status **[P]**.

8.4 Open Problems

1. **Derive the powers 12 and 13** from the 5D EDC action
2. **Derive the factor $128\pi^2$** from geometry
3. **Derive r_e from first principles** — is it fundamental or emergent?
4. **Extend to other particles** — do protons, neutrons follow the same pattern?
5. **Strong-field predictions** — where does EDC differ from GR?
6. **Cosmological implications** — dark matter, dark energy, Hubble tension

8.5 What Must Never Be Used

Forbidden Formulas

- $\hbar = \sigma R_\xi^3/c$ — **WRONG** (off by $10^{10}!$)
- $G = c^2/(4\pi\sigma)$ — **WRONG** (off by $10^8!$)
- $\rho_{\text{Plenum}} \sim 10^{97} \text{ kg/m}^3$ — **UNRELIABLE**
- M for vortex energy — **USE $\mathcal{M}_v$ INSTEAD**

Appendix A

Notation Reference

Symbol	Meaning	Dimensions
M	Gravitational source mass	kg
$\mathcal{M}_v$	Vortex equivalent mass ($\sigma r_e^2/c^2$)	kg
$\mathcal{E}_v$	Vortex elastic energy (σr_e^2)	J
m_e	Electron mass	kg
r_e	Classical electron radius	m
R_ξ	Compact dimension	m
r_{core}	Core radius (GM/c^2)	m
r_s	Schwarzschild radius ($2GM/c^2$)	m
σ	Membrane tension	$\text{J/m}^2 = \text{kg/s}^2$
ν_{bulk}	Kinematic viscosity	m^2/s
p_∞	Plenum pressure at infinity	Pa

Table A.1: Complete symbol reference.

Appendix B

Dimensional Analysis Reference

Quantity	Dimensions
Length	$[L]$
Mass	$[M]$
Time	$[T]$
Velocity	$[LT^{-1}]$
Acceleration	$[LT^{-2}]$
Force	$[MLT^{-2}]$
Energy	$[ML^2T^{-2}]$
Pressure	$[ML^{-1}T^{-2}]$
Surface tension	$[MT^{-2}]$
Viscosity (dynamic)	$[ML^{-1}T^{-1}]$
Viscosity (kinematic)	$[L^2T^{-1}]$
Gravitational constant	$[L^3M^{-1}T^{-2}]$
Planck constant	$[ML^2T^{-1}]$

Table B.1: Standard dimensional analysis reference.

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